SIDDHANT PODDAR

siddhantpoddar48@gmail.com | +91-8811922513 | linkedin.com/in/siddhant-poddar | github.com/sidpoddar07 | Bangalore, India

PROFESSIONAL SUMMARY

Computer Science (Cybersecurity) undergraduate with strong foundations in application security, full-stack web engineering, and scalable backend architectures. Experienced in building zero-trust API security gateways, interactive monitoring dashboards, and database-driven systems using Python, JavaScript/React, Java, and FastAPI. Proficient in secure software development (OWASP Top 10), cryptographic protocols, and system automation, with a proven ability to deliver reliable end-to-end applications.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, C, SQL, Bash / Shell Scripting
- **Frontend & Web Development:** React, HTML5, CSS3, TailwindCSS, Vite, RESTful API Integration, Responsive UI Design
- **Backend & Architecture:** FastAPI, Asynchronous I/O (asyncio), Pydantic v2, Object-Oriented Design, Microservices & System Architecture
- **Cybersecurity & Cryptography:** Application Security (OWASP Top 10), Zero-Trust Architecture, Cryptography (AES, RSA, HMAC-SHA256), RBAC, Network Security (TCP/IP), Packet Analysis
- **Tools, Cloud & Databases:** Git, GitHub, Wireshark, Burp Suite, Nmap, Postman, AWS Fundamentals (IAM, S3, VPC), MySQL, PostgreSQL, In-Memory Data Structures

FEATURED PROJECTS

AegisPay-AI — Zero-Trust Security Mesh & Payment Firewall for AI Agents `FLAGSHIP`

GitHub: **github.com/sidpoddar07/AegisPay-AI** | *Tech Stack: Python, FastAPI, Pydantic, React, Vite, Cryptography, Pytest*

- Engineered a high-throughput, in-flight Zero-Trust security gateway that intercepts autonomous AI agent payment requests before touching payment gateway APIs (e.g., Razorpay).
- Designed an **In-Flight Cascaded Architecture** achieving a mean latency of `0.036 ms` (< 1ms SLA) by combining pre-compiled regex automata, Shannon entropy obfuscation detection, and bidirectional intent divergence algorithms.
- Implemented a **Cryptographic Proof-of-Intent (PoI) Token** using time-bound (60s TTL) HMAC-SHA256 to eliminate in-flight Man-In-The-Middle (MITM) parameter tampering and amount inflation attacks.
- Developed a 3-tier composite risk matrix with an adaptive **`REVIEW` band** (Risk 30–74) to trigger Step-Up 2FA challenges, reducing false-positive checkout drops.
- Built an automated regulatory compliance engine that synthesizes **RBI-AML/PMLA-compliant Suspicious Activity Reports (SAR)** sealed with SHA-256 tamper-evident integrity hashes.
- Achieved **100% test coverage (32/32 passing tests)** and created a real-time Cyber SOC Dashboard with live penetration attack simulation.

Smart Attendance Management System

Tech Stack: Python, Web Technologies, MySQL, RBAC, REST APIs

- Developed a secure web-based attendance automation platform reducing manual operational overhead by 80%.
- Implemented robust session-based authentication, Role-Based Access Control (RBAC), and parameterized SQL queries to safeguard student data against injection attacks.
- Architected a centralized reporting dashboard with optimized indexing for high-speed attendance record retrieval.

Linux Automation & Infrastructure Toolkit

Tech Stack: Bash, Shell Scripting, Linux System Administration, Cron

- Engineered modular Bash automation utilities for automated log rotation, resource threshold alerting, and background task scheduling.
- Implemented defensive scripting patterns with automated error-handling traps, permission auditing, and secure file integrity checks.

EDUCATION

B.Tech in Computer Science (Cyber Security) | *Dayananda Sagar University, Bangalore* — **2023 – 2027**

Higher Secondary Certificate (12th) | *Kendriya Vidyalaya* — **2020 – 2022 | Score: 74.20%**

CERTIFICATIONS & TRAININGS

• **IBM Ethical Hacking Principles** | • **Fundamentals of Information Security** | • **Fundamentals of Cryptography**

• **CompTIA Security+** *(In Progress)* | • **AWS Cloud Practitioner** *(In Progress)*